

6.11.7.4. Data Field

This field SHALL indicate Optional app-specific data.

6.12. Content App Observer Cluster

This cluster provides an interface for sending targeted commands to an Observer of a Content App on a Video Player device such as a Streaming Media Player, Smart TV or Smart Screen.

The cluster server for Content App Observer is implemented by an endpoint that communicates with a Content App, such as a Casting Video Client.

The cluster client for Content App Observer is implemented by a Content App endpoint.

A Content App is informed of the `NodeId` of an Observer when a binding is set on the Content App. For a Content App Platform, the binding is set by the platform when a `CastingVideoClient` is granted access to the Content App, and the `CastingVideoClient` supports the Content App Observer cluster. The Content App can then send the `ContentAppMessage` to the Observer (server cluster), and the Observer responds with a `ContentAppMessageResponse`.

The `Data` and `EncodingHint` fields of the `ContentAppMessage` and `ContentAppMessageResponse` contain content app-specific values, the format and interpretation of which is defined by the Content App vendor, analogous to the custom message features offered by other popular casting protocols. Standardized cluster and commands are used here rather than manufacturer-specific cluster and commands because of the role that the Content App Platform plays in creating the ACLs and Bindings on both sides of the communication between the Content App Observer endpoint and the Content App endpoint.

By using standard cluster and commands:

1. The Content App Platform is able to easily determine that a binding is needed on the Content App endpoint because it can recognize the Content App Observer cluster implemented by a client node.
2. The Content App Platform is able to easily identify commands that are allowed to be sent by the Content App to a client node because those commands use the Content App Observer cluster.
3. The Content App is able to easily determine that a node supports the Content App Observer cluster because it has received a binding which specifies the Content App Observer cluster.
4. The Casting Video Client is able to support a single cluster for receiving commands from any Content App and does not need to explicitly list every Content App it understands.

A Content App Observer SHOULD ignore the `Data` and `EncodingHint` field values in commands from a Content App it does not recognize. A Content App SHOULD ignore the `Data` field values in responses when the `EncodingHint` value is blank or not recognized.

6.12.1. Revision History

The global `ClusterRevision` attribute value SHALL be the highest revision number in the table below.